2.8, 6. DETERMINE IF $\begin{pmatrix} -4 \\ 10 \\ -7 \\ -5 \end{pmatrix}$ IS IN THE SPAN

OF $\begin{pmatrix} 1 \\ -2 \\ 4 \\ 3 \end{pmatrix}$, $\begin{pmatrix} 4 \\ -7 \\ 9 \\ 7 \end{pmatrix}$, AND $\begin{pmatrix} 5 \\ -8 \\ 6 \\ 5 \end{pmatrix}$. IN OTHER WORDS,

ARE THERE $a_1, a_2, a_3 \in \mathbb{R}$ S.T.

$$a_1\begin{pmatrix} 1 \\ -2 \\ 4 \\ 3 \end{pmatrix} + a_2\begin{pmatrix} 4 \\ -7 \\ 9 \\ 7 \end{pmatrix} + a_3\begin{pmatrix} 5 \\ -8 \\ 6 \\ 5 \end{pmatrix} = \begin{pmatrix} -4 \\ 10 \\ -7 \\ -5 \end{pmatrix}, \quad \text{I.E.,}$$

DOES $\begin{pmatrix} 1 & 4 & 5 \\ -2 & -7 & -8 \\ 4 & 9 & 6 \\ 3 & 7 & 5 \end{pmatrix}\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} -4 \\ 10 \\ -7 \\ -5 \end{pmatrix}$ HAVE A SOL'N?

ROW REDUCE THE AUGMENTED MATRIX

$$\begin{pmatrix} 1 & 4 & 5 & -4 \\ -2 & -7 & -8 & 10 \\ 4 & 9 & 6 & -7 \\ 3 & 7 & 5 & -5 \end{pmatrix} \sim \begin{pmatrix} 1 & 4 & 5 & -4 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 23 \\ 0 & 0 & 0 & 0 \end{pmatrix} \leftarrow 0a_1 + 0a_2 + 0a_3 = 23$$

NO SUCH (a_1, a_2, a_3)

ANSWER: NO.

2.8, 32. FIND A BASIS FOR COL(A) AND NUL(A).

$$A = \begin{pmatrix} -3 & 9 & -2 & -7 \\ 2 & -6 & 4 & 8 \\ 3 & -9 & -2 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & -3 & 6 & 9 \\ 0 & 0 & 4 & 5 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

THM 13: THE PIVOT COLUMNS of A FORM A

BASIS FOR COL(A).

BY THM 13, $\begin{pmatrix} -3 \\ 2 \\ 3 \end{pmatrix}$ AND $\begin{pmatrix} -2 \\ 4 \\ -2 \end{pmatrix}$ FORM A BASIS

FOR COL(A).

$x \in \mathrm{Nul}(A) \iff x \in \mathrm{Nul} \begin{pmatrix} 1 & -3 & 6 & 9 \\ 0 & 0 & 4 & 5 \\ 0 & 0 & 0 & 0 \end{pmatrix}$.

$\iff x_1 - 3x_2 + 6x_3 + 9x_4 = 0$ AND

$4x_3 + 5x_4 = 0$.

$\iff x_4 = -\frac{4}{5}x_3$ AND $x_1 - 3x_2 + 6x_3 - \frac{36}{5}x_3 = 0$,

$\iff x_3 = \frac{5}{6}x_1 - \frac{5}{2}x_2$. AND $x_4 = -\frac{2}{3}x_1 + 2x_2$

THUS $X \in NUL(A)$ IF AND ONLY IF

$$X = \left(x_1, x_2, \frac{5}{6}x_1 - \frac{5}{2}x_2, -\frac{2}{3}x_1 + 2x_2\right)$$

$$= x_1 \begin{pmatrix} 1 \\ 0 \\ \frac{5}{6} \\ -\frac{2}{3} \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ 1 \\ -\frac{5}{2} \\ 2 \end{pmatrix}$$ FOR SOME $x_1, x_2 \in \mathbb{R}$.

SO $\left\{ \begin{pmatrix} 1 \\ 0 \\ \frac{5}{6} \\ -\frac{2}{3} \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -\frac{5}{2} \\ 2 \end{pmatrix} \right\}$ FORMS A BASIS FOR

$NUL(A)$.